



Andy Kauth (Attempt 1 of

1)

Test Information

Current Grade 95.00000 out of 100 points
Grade based on Last Evaluated Attempt

Status Completed

Attempt Score 95 out of 100 points

Time Elapsed 169 hours, 10 minutes

Started Date 11/24/25 5:02 PM

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Submitted Date 12/1/25 6:12 PM

Due Date 12/1/25 11:59 PM

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Instructions

QUESTION 1: CALCULATED NUMERIC

25

out of 25 points

Refer to page 334 of Chapter 7 that is uploaded on the UIC Blackboard in the Course Textbook content area.

1. Identify the error(s) in the calculation of the Dew Point pressure
2. Propose a detailed algorithm for the correct calculation of the Dew Point pressure.



3. Carry out the calculation and enter your result in **bar** in the textbox below. Round your numerical answer into **two decimal digits**.

Given Answer:  3.69

Correct Answer:  3.7

Answer range +/- 0 (3.7 - 3.7)

QUESTION 2: FILE RESPONSE

70

out of 75 points

CHE 301 Take Home Midterm 3 Report

When it comes to the proposed solution for the dew point pressure in chapter 7 of the textbook, on page 334, there are a couple of critical errors that were made which resulted in the incorrect value of the dew point pressure.

One major error that was made was using activity coefficients that were evaluated at the wrong composition. The activity coefficients that were used were computed earlier, the ones for the bubble point composition and the overall composition, when in reality the activity coefficients that should have been used were the ones for the liquid composition at the dew point. However, the liquid coefficients at the dew point depend on the dew point pressure itself, so y_1 must be evaluated at the liquid composition that results in that pressure, making this an implicit problem.

Another major error that was made was, as somewhat stated above, treating this problem as an explicit problem, when in reality the problem is implicit. The proper dew-point equation contains $y_i(x)$ and x_i , which both depend on P_{dew} . So you cannot simply plug in values for y_i that were computed for a different composition to get the correct value for the dew point pressure, unless it has been shown that y is completely independent of x .

At the dew point the vapor composition equals the given vapor composition $y_i = z_i$. Raoult's-type equilibrium gives,

Rearrange to get the liquid composition in terms of P_{dew} and γ_i .

Enforce molar summation $\sum_i x_i = 1$.

Or equivalently,

It is important to note that on the rhs is a function of the liquid composition x_i and x itself depends on P_{dew} . So the above equation is an implicit equation for P_{dew} and must be solved iteratively. You cannot evaluate the rhs using γ_i computed at an unrelated composition and expect a proper exact solution.

We then use the corrected formula,

Through the CHE Calculator, running an iterative flash calculation to solve for the correct values and equating the summation, we get P_{dew} to be,